

myNRC-OCR: A Hybrid OCR Framework for Handwritten Myanmar NRCs Integrating YOLOv8, Tesseract LSTM, TrOCR, and Post-OCR Correction

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Abstract—Handwritten OCR for Myanmar script is hindered by complex orthography, character stacking, and limited annotated data. We propose myNRC-OCR, a hybrid framework for robust text extraction from Myanmar National Registration Cards (NRCs). The pipeline utilizes YOLOv8 for field localization, followed by a comparative recognition study between Tesseract LSTM and Transformer-based TrOCR. Analysis reveals a shift from segmentation-based failures in LSTMs to generative hallucinations in Transformers. To mitigate these, we introduce a 3-tier post-processing module using syllable tokenization and rule-based normalization to enforce linguistic constraints. Results show that while Tesseract is computationally efficient, the fine-tuned TrOCR model achieves a superior Character Error Rate (CER) of 2.46%, effectively managing high-variance handwriting. This work provides a scalable methodology for digitizing low-resource Southeast Asian scripts and establishes a foundation for future Myanmar linguistic research.

Index Terms—Optical Character Recognition, Myanmar Script, Deep Learning, KYC Automation, YOLOv8, TrOCR, Low-Resource Languages

I. Introduction

Digitizing handwritten documents in low-resource linguistic environments remains a significant challenge in document image analysis. For the Myanmar script, this is compounded by circular orthography, non-linear character stacking, and complex diacritics. National Registration Cards (NRCs) represent a unique intersection of these hurdles, featuring handwritten entries superimposed on high-entropy, decorative backgrounds that disrupt standard binarization and segmentation pipelines.

Traditional OCR frameworks often struggle with the structural variability of handwritten Burmese. While general-purpose detectors like PaddleOCR excel at East Asian scripts, they frequently fail to capture the phonetic dependencies of stacked consonants. This research utilizes **Tesseract LSTM** as a robust baseline for processing

Indic-derived scripts via connectionist temporal classification (CTC). To address the high variance of cursive handwriting, we further explore the **Vision Transformer (ViT)** architecture via **TrOCR**. By treating OCR as a sequence-to-sequence task, we bypass explicit character segmentation—a primary failure point in traditional engines.

However, neural outputs in low-resource settings are prone to “hallucinations” and orthographic inconsistencies. To address this, we propose **myNRC-OCR**, a hybrid framework contributing:

- **Localized Feature Extraction:** An optimized **YOLOv8** implementation for precise region-of-interest (ROI) detection in complex layouts.
- **Comparative Recognition Pipeline:** A dual-path approach evaluating the stability of **Tesseract LSTM** against the generative flexibility of **TrOCR**.
- **Linguistic Refinement:** A 3-tier post-processing module employing syllable tokenization and fuzzy matching to rectify orthographic errors.

This work offers a methodological blueprint for Southeast Asian script recognition, providing insights into error mode transitions between recurrent and transformer-based architectures in low-data regimes.

II. Image Dataset Preparation

A. Data Acquisition and Preprocessing Pipeline

The dataset was constructed from high-resolution NRC images retrieved via **AWS** from a production environment, encompassing diverse real-world noise and lighting conditions. Key textual regions—Name, Father’s Name, DOB, and NRC ID—were isolated using a **YOLOv8** object detection model. Following localization, crops underwent grayscale conversion and **Otsu thresholding** to remove decorative background patterns and enhance stroke contrast for OCR engines.

Data was collected with explicit user consent for system improvement. All processing protocols strictly adhere to regional data protection guidelines, ensuring that personally identifiable information (PII) is encrypted and isolated to maintain privacy and security.

B. Object Detection Dataset

A YOLOv8 pipeline isolates four key regions: Name, Father Name, NRC No, and Date of Birth. While the dataset comprises 369 images (see Table I), each sample contains multiple class instances, providing sufficient feature density for learning spatial relationships. Empirical testing on unseen, diverse NRC samples confirmed high localization accuracy without significant regression, indicating that the model generalizes well to real-world variations despite the constrained image count.

TABLE I
Dataset Distribution for YOLOv8 Field Detection Model

Split	Field Detection (Images)
Train	258
Valid	74
Test	37
Total	369

C. Dataset Distribution and Availability

Two distinct datasets meet specific model requirements. The Tesseract dataset uses a field-specific organization (Name, NRC ID, DOB) for targeted optimization. The TrOCR dataset employs a unified image-text pairing strategy partitioned into training, validation, and testing subsets for sequence learning. A synthetic sample is available for reproducibility on Hugging Face at huggingface.co/datasets/PyaeLinn/myNRC-OCR_Sample-Dataset.

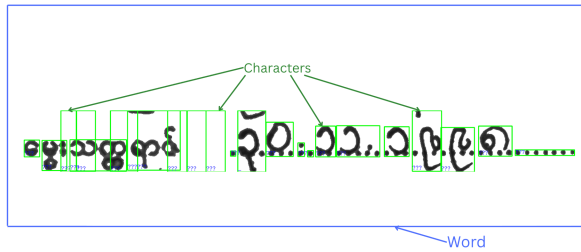


Fig. 1. Character-level Segmentation of the Myanmar phrase "မွေးသက္ကရာဇ် ၁၀.၁၁.၁၉၉၈" (Date of Birth: 30.11.1998) in which Characters in Green and Word in Blue boxes

III. Methodologies

The **myNRC-OCR** framework implements a three-stage pipeline optimized for the Myanmar script:

- 1) **Field Localization:** YOLOv8 identifies and crops specific NRC fields (ID, Name, Father's Name, DOB) to isolate textual regions from complex document backgrounds.

TABLE II
Dataset Distribution for Fine-tuning Tesseract OCR

Field	Images	Characters
Name (Name + Father Name)	15,602	~374,448
NRC ID	9,460	~236,500
Date of Birth	5,976	~137,448
Total	31,038	~748,396

TABLE III
Dataset Distribution for TrOCR Model

Split	Images	Percentage
Train	21,985	70%
Valid	6,281	20%
Test	3,141	10%
Total	31,407	100%

- 2) **Hybrid Recognition:** Two complementary architectures are fine-tuned on the localized crops:

- **Tesseract LSTM:** Optimized via character-aligned .box files and Otsu-binarized images to provide a robust, lightweight baseline.
- **TrOCR:** A transformer-based vision-language model trained on augmented image-text pairs to handle high-variance handwriting through global attention.

- 3) **Post-OCR Correction:** A final module utilizes rule-based logic and fuzzy matching to enforce NRC-specific linguistic and structural syntax.

A. Fine-Tuning Tesseract OCR for Myanmar NRC

1) **Field Extraction and Ground Truth:** The pipeline begins with a YOLOv8 model localizing and cropping key fields (e.g., Date of Birth) from high-resolution NRC images. These crops are preprocessed using OpenCV (Otsu thresholding) to generate clean feature maps. Ground truth labels are normalized to Unicode NFC and paired with .gt.txt files to supervise the transcription layer.

2) **Training Pipeline:** To bridge the gap between visual features and recurrent layers, we generate .box files mapping Myanmar characters to pixel coordinates. These are compiled into .lstmf files, enabling the Deep Bidirectional LSTM to learn the sequential dependencies of the script, as illustrated in the architectural layers of Fig. 2.

3) **Fine-Tuning:** Using `mya.traineddata` as a base, we iteratively fine-tune the recurrent and transcription layers. The model learns to map complex handwritten feature sequences into accurate predicted strings, resulting in a specialized engine optimized for the orthographic challenges of NRC documents. The complete end-to-end flow from input image to predicted sequence is detailed in Fig. 2.

B. Fine-Tuning TrOCR for Myanmar NRC

1) **Dataset and Normalization:** A dataset of 31,407 cropped images (Names, NRC IDs, DOB) was curated and stratified

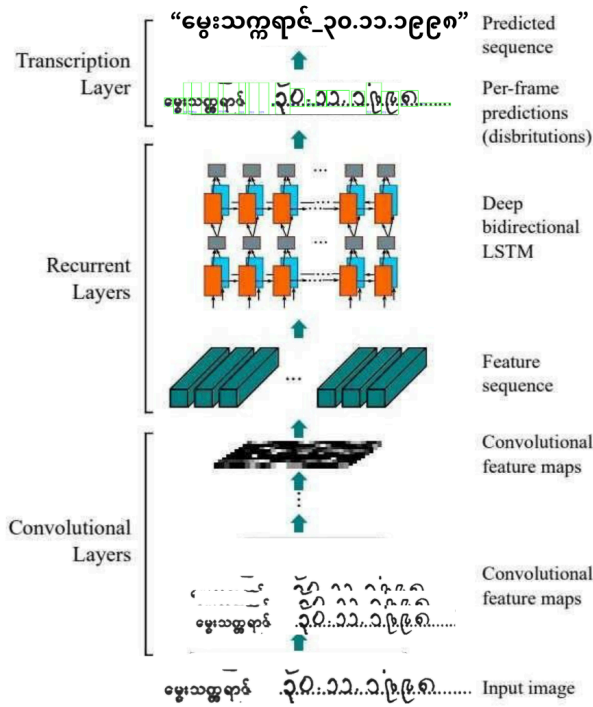


Fig. 2. Architecture of the Fine-Tuned Tesseract OCR Model for Myanmar NRCs (Adapted from: Structure of the Tesseract OCR, ResearchGate Fig. 4, accessed 24 Dec 2025).

(Table III). We applied a normalization pipeline using unicodedata and regex to ensure linguistic consistency: labels were standardized to Unicode NFC, zero-width characters (U+200B/U+200C) were stripped, and artifacts like underscores or numeric suffixes were cleaned.

2) *Data Augmentation*: To generalize across irregular handwriting, an *Albumentations* pipeline applied:

- **Geometric**: *Grid Distortion* and *Elastic Transformations* to simulate local stroke warping.
- **Photometric**: *RGB Shifts*, *Gaussian Noise*, and *Blur* to mimic low-quality mobile captures.
- **Orientation**: Rotations within $\pm 5^\circ$ for scanning alignment.

3) *Architecture and Optimization*: The model pairs a ViT encoder (12 layers, 768 units) with a SEA-LION BERT decoder (12 layers, 1024 units), totaling **806M parameters** (Fig. 3). The tokenizer was expanded to 256,000 tokens for Myanmar script coverage. For training on a single NVIDIA L4 (24GB VRAM), we utilized FP16 mixed-precision, gradient checkpointing, and a gradient accumulation step of 4.

4) *Training Configuration*: Fine-tuning ran for 30 epochs with a $1e-5$ learning rate, 0.01 weight decay, and a cosine schedule (10% warmup). We used a per-device batch size of 4, optimized for stable convergence within hardware constraints.

C. Post-OCR Correction

A post-processing module normalizes and validates OCR outputs against NRC standards to handle handwriting variability and complex scripts.

1) NRC ID and DOB Normalization:

- **NRC ID**: Township codes are verified against a JSON registry and corrected when the Levenshtein distance is 1 with a unique match. Serial numbers are normalized to အမှတ်_xx/yyy(z)dddddd.
- **DOB**: Digits are converted to ASCII, invalid values clamped, and reconstructed as မွေးသက္ကရာဇ်_dd.mm.yyyy in Myanmar script.

2) *Hybrid Name Correction*: Open-vocabulary names are syllabified using *sylbreak.py* and corrected via a 3-tier pipeline based on N-gram confidence:

- **Tier 1** (> 0.8): Fast **SymSpell** lookup to avoid over-correction.
- **Tier 2** (0.5–0.8): SymSpell candidates ranked by N-gram frequency.
- **Tier 3** (< 0.5): Aggressive sub-word fuzzy matching and structural constraint enforcement for heavily corrupted syllables.

IV. Experiments

A. Optical Model Training with Tesseract

Tesseract was fine-tuned exclusively on handwritten Myanmar NRC data for KYC robustness.

- **Field-Specific Training**: Separate models were trained for (i) Names (Name + Father's Name), (ii) NRC IDs, and (iii) Dates of Birth.
- **Training Iterations**: Name model—**9,200** (high stroke variability); NRC ID—**8,191** (digits and structure); DOB—**6,318** (numeric-dominant).
- **Training Stability**: Checkpointed iterative training was used to monitor convergence and reduce overfitting to individual handwriting styles.

B. Optical Model Training with TrOCR

Fine-tuned on 31,407 image-text pairs over 30 epochs, the TrOCR model achieved convergence by epoch 18. Unlike Tesseract's struggle with irregular Burmese handwriting spacing, TrOCR's attention mechanism implicitly learned vertical stacking and global stroke patterns without explicit character segmentation.

C. OCR Error Analysis in Tesseract

As summarized in Table IV, Tesseract exhibits recurring failure modes stemming from LSTM sequence instability and script complexity.

Prefix and Structural Bias: Strong memorization of frequent patterns leads to duplicated or fragmented prefixes in Name (ဖခင်အမည်), DOB (မွေးသက္ကရာဇ်), and ID (အမှတ်) fields (Cases 2, 5, 6). This structural bias often results in "hallucinated" repetitions when the model encounters low-contrast handwriting.

Orthographic and Diacritic Errors: The model struggles with Myanmar's complex stacking and diacritics, frequently inserting extra marks (e.g., ဝး) or omitting characters in

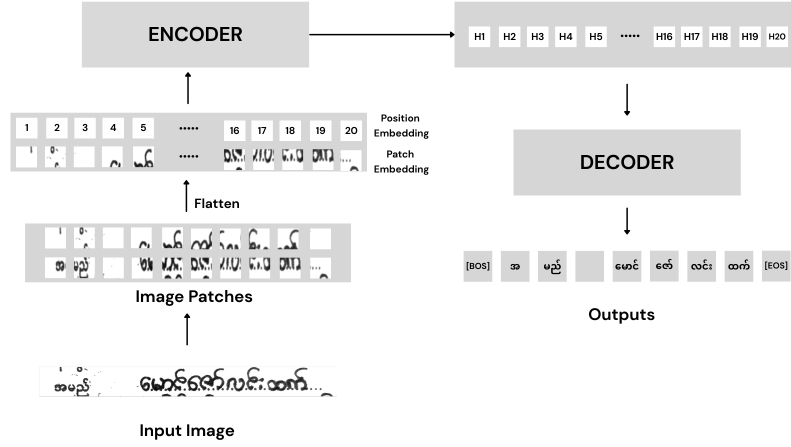


Fig. 3. **Fine-Tuned TrOCR Architecture.** The ViT encoder processes 16×16 patches; the decoder uses cross-attention for autoregressive Myanmar sequence generation.

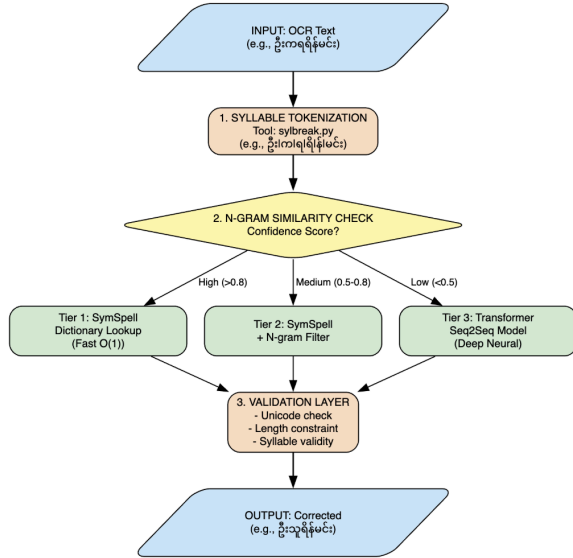


Fig. 4. Overview of the 3-Tier Hybrid Post-Processing Pipeline. The system routes OCR outputs through SymSpell or a Transformer Seq2Seq model based on confidence scores.

stacked consonants and foreign phonetic structures (Cases 1, 3, 4).

Visual Ambiguity: Handwritten variations cause misclassification of visually similar characters, such as the confusion between မ and ဗ (Case 7), highlighting the limitations of the feature extraction layer under weak stroke separation.

D. OCR Error Analysis in TrOCR

TrOCR achieved superior accuracy (2.46% CER) by shifting error profiles from segmentation-based failures to generative hallucinations.

TABLE IV
Representative Tesseract OCR Errors in Myanmar NRCs

Case	Ground Truth	OCR Error Output
1	ဦးအောင်ဌေးဦး	ဦးအောင်ဌေး
2	ဖခင်အမည်_စောဘေဆင်း	အမည်_ဖအမည်_မ်စ
3	အဗ္ဗဒ္ဒဆတ္တိရ်	အမည်_်
4	မွေးသက္ကရာဇ်_၁၁.၇.၁၉၉၄	မွေးသက္ကရာဇ်_၁၁.၇.၁၉၉၄
5	၁၇.၁၁.၂၀၀၁	မွေးသက္ကရာဇ်_မွေးသက္ကရာဇ်_၂၀၀၁
6	အမှတ်_စ-စလန...	အမှတ်_စ- အမှတ်စဉ်...
7	မကန(နိုင်)	ဗကန(နိုင်)

Background Robustness: Unlike Tesseract’s susceptibility to holographic artifacts, TrOCR’s attention mechanism isolates ink density from non-semantic noise. As shown in Fig. 5, the model maintains high invariance to circular NRC patterns that traditional engines often misidentify as characters.

Seq2Seq Hallucinations: TrOCR’s limitations stem from prioritizing contextual probability over visual evidence in low-resource settings:

- **Sequence Truncation:** Performance degrades on long or compound names (> 5 syllables) due to positional embedding constraints.
- **Auto-Completion Bias:** The decoder may “hallucinate” common N-grams (e.g., frequent names) over faint visual evidence.
- **Orthographic Forcing:** The model occasionally inserts omitted tone marks (ံ, ့း) to enforce standard script rules, even if absent in informal handwriting.

These reliability issues underscore the necessity of the Tier-3 Post-Processing pipeline for structural verification.

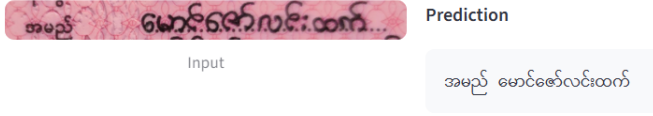


Fig. 5. TrOCR robustness to background artifacts.

E. Post-OCR Experiments

The post-processing framework implements a two-tier correction strategy to handle remaining sequence errors:

Statistical Correction: We utilized **SymSpell** (max edit distance 4) integrated with Character N-gram ranking. Empirical evaluation identified Trigrams ($N = 3$) as the optimal configuration, providing superior contextual balance compared to less granular bigrams or overly sparse 4-grams.

Heuristic Fallback: For severe corruptions where N-gram confidence falls below threshold (Tier 3), an aggressive Levenshtein-based alignment is triggered. This module forces synchronization with a valid Myanmar syllable dictionary derived from NRC nomenclature. This approach prevents orthographically invalid sequences while maintaining lower computational overhead than additional neural refinement.

F. System Configuration

Experiments utilized decoupled cloud and local environments matched to model-specific demands. The memory-intensive **TrOCR** was fine-tuned on *Google Colab Pro+* using an **NVIDIA L4 GPU (22GB VRAM)** and 53GB RAM. Conversely, **Tesseract LSTM** was trained in **CPU-only mode** on a compute-optimized instance, leveraging high-capacity RAM for glyph feature caching. The **Post-OCR Correction** module—comprising SymSpell and N-gram ranking—was executed locally on an Intel i5 with 8GB RAM, reflecting the pipeline’s minimal computational overhead and suitability for edge deployment.

G. Evaluation Metrics

System performance is quantified using character and word-level metrics to evaluate OCR accuracy and post-processing efficacy.

1) *CER, WER, and CHRF++*: Character Error Rate (CER) and Word Error Rate (WER) calculate the percentage of errors relative to the ground truth length N : $CER/WER = \frac{S+D+I}{N} \times 100$, where S, D, I are substitutions, deletions, and insertions. CER is the primary metric for training, while WER assesses full-field extraction accuracy.

For TrOCR, we incorporate **CHRF++** to better capture Myanmar script’s morphological complexity: $CHRF++ = \frac{(1+\beta^2)(chr_p \cdot chr_r)}{\beta^2 \cdot chr_p + chr_r}$, balancing N-gram precision (chr_p) and recall (chr_r).

2) *Correction Efficacy*: Post-OCR module impact is measured by the absolute reduction in error rates: $\Delta Error = Error_{raw} - Error_{corrected}$. A positive Δ validates the normalization and structural enforcement logic of the Tier-3 pipeline.

H. Experimental Results

1) Field Localization and Segmentation (YOLOv8):

Accurate field extraction underpins the pipeline. As shown in Table V (Right), YOLOv8 achieved **0.83 precision** and **0.79 mAP@50**, demonstrating robust field localization despite complex backgrounds. Although validation loss exceeded training loss, a high recall (0.82) ensured reliable crop generation for OCR.

TABLE V
Field Localization Performance (YOLOv8)

Metric	Value
Precision	0.832
Recall	0.820
mAP@50	0.791
mAP@50-95	0.428
Train Box Loss	1.251
Val Box Loss	1.487

2) *OCR Comparison: Tesseract vs. TrOCR*: Recognition performance was compared between Tesseract-LSTM and Transformer-based TrOCR (Table VI, Left). **TrOCR** achieved a substantially lower average CER (**2.46%**) than Tesseract (13.39%), with the largest gain in *Name* fields (+20.2%). While Tesseract offers lower latency, TrOCR provides the accuracy required for dependable KYC verification.

TABLE VI
OCR Performance Comparison: Tesseract vs. TrOCR (CER %)

Field	Tesseract	TrOCR	Improvement
Name	23.32	3.12	20.20
NRC ID	10.99	2.41	8.58
Date of Birth	5.85	1.85	4.00
Average	13.39	2.46	10.93

3) *Post-OCR Correction Evaluation*: We evaluated the correction module’s internal stability and inference capabilities. To isolate and quantify the contribution of each stage within our hybrid correction pipeline, we conducted an ablation study tracking the Character Error Rate (CER) across the validation set. Starting from the raw TrOCR baseline CER of 2.46%, Tier 1 (Fast SymSpell Lookup) efficiently resolved simple misspellings, reducing the CER to 2.13%. Tier 2 (N-gram filtering) further disambiguated contextually similar syllables, yielding a CER of 1.55%. Finally, Tier 3 (Heuristic Fallback) successfully reconstructed heavily corrupted tokens, achieving the final optimized performance of 1.44%. This step-by-step reduction validates the necessity of the tiered approach.

4) *Significance and Latency Analysis*: To validate the performance gains, we conducted paired bootstrap resampling ($B = 1000$) on the test set. The CER improvement of TrOCR over Tesseract was found to be statistically significant ($p < 0.001$) with a 95% confidence interval of [2.34%, 2.58%].

Furthermore, we benchmarked inference latency to assess suitability for real-time applications (Table VII). On an NVIDIA L4 GPU, the full TrOCR pipeline averages 275ms

per card, whereas Tesseract (CPU) achieves 105ms. While Tesseract is approximately $2.6\times$ faster, the 275ms latency of TrOCR remains well within the acceptable threshold for automated verification, justifying the trade-off for its superior accuracy.

TABLE VII
Inference Latency Comparison (Average per Card)

Component	Tesseract (CPU)	TrOCR (GPU)
Field Localization (YOLOv8)	15ms	15ms
OCR Recognition	80ms	250ms
Post-Processing	10ms	10ms
Total Pipeline	105ms	275ms

V. Conclusion and Future Work

This work presents an end-to-end OCR pipeline for Myanmar NRC text extraction, combining three key components: a fine-tuned Tesseract model, a transformer-based TrOCR model, and a post-OCR correction module. YOLOv8-based field segmentation ensures clean inputs, while Tesseract provides strong character-level recognition and TrOCR offers robustness for handwritten and complex Myanmar script. The post-correction stage enforces NRC formatting and fixes common Myanmar script errors, resulting in a reliable system for extracting ID number, name, father’s name, and date of birth. Together, these approaches demonstrate that a hybrid OCR strategy significantly improves accuracy for low-resource Myanmar document digitization.

Future enhancements include expanding the dataset size for all NRC fields, improving model robustness through continuous fine-tuning, and strengthening post-OCR correction using both rule-based and learning-based techniques. Deployment-focused improvements—such as adopting MLOps pipelines, automating retraining schedules, and preparing scalable inference services—will support real-world fintech KYC integration. These steps aim to further increase accuracy, maintain long-term reliability, and enable seamless deployment of a production-ready Myanmar NRC OCR system.

A. System Availability

To facilitate reproducibility and demonstrate real-world applicability, we provide the following resources:

- **Code and Synthetic Data:** The open-source frontend, API wrapper, sanitized pre-processing and training scripts are available at the `mynrc-ocr` repository under the username `zawlinnhtet03` on GitHub (github.com/zawlinnhtet03/mynrc-ocr).
- **Live Demonstration:** A deployed version of the myNRC-OCR system is hosted on the Streamlit platform (`dingerkyc.streamlit.app`). Note that due to data privacy regulations regarding National Registration Cards, access is restricted; credentials for academic verification may be requested from the corresponding author.

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